

Emmanuel Suárez Acevedo

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Work Experience

Graduate Researcher

Cornell University

New York, NY

August 2024 – Present

- * Developing formal verification techniques for randomized concurrent systems with fair scheduling guarantees, targeting correctness of distributed consensus protocols
- * Designed programming language for automated verification of quantitative network behavior (e.g., *is all traffic delivered within 9ms?*) and presented paper at PLDI 2026

Graduate Researcher

University of Pennsylvania

Philadelphia, PA

August 2022 – April 2024

- * Developed type system for statically tracking computational effects and resource demands, proving dead code elimination preserves program behavior (OOPSLA 2024)

Senior Software Engineer

Strava

San Francisco, CA

August 2019 – July 2022

- * Led platform-wide migration of core activities data model (1B+ records) into a centralized microservice to unlock new sport types (trail running, MTB, etc.)
- * Investigated malformed route data, identifying/fixing root causes and then shipping Apache Spark backfill job and request hooks to remediate affected data
- * Used Apache Kafka and Spark to build geospatial data pipeline powering Strava Metro, a platform used by transportation agencies worldwide for urban planning
- * Developed Scala microservice using Finatra, Apache Thrift, and Cassandra to serve millions of static map rendering requests across iOS, Android, and web

Education

Cornell University Ann S. Bowers College of Computing and Information Science

PhD in Computer Science

New York, NY

August 2024 – Present

University of Pennsylvania School of Engineering and Applied Science

MSE in Computer and Information Science

Philadelphia, PA

August 2022 – May 2024

University of Pennsylvania School of Engineering and Applied Science

BSE with Honors in Computer and Information Science

Philadelphia, PA

August 2015 – May 2019

Skills

Programming Languages: OCaml, Scala, Python, Haskell, Java, Ruby, C/C++

Software Engineering: Git, Rails, Apache Spark, Apache Kafka, Apache Thrift, Cassandra, MySQL

Formal Verification: Agda, Rocq, Lean

Publications

Weighted NetKAT: A Programming Language for Quantitative Network Verification

PLDI 2026

Emmanuel Suárez Acevedo, Tiago Ferreira, Kevin Batz, Oliver Bøving, Nate Foster, and Alexandra Silva

Effects and Coeffects in Call-By-Push-Value

OOPSLA 2024

Cassia Torczon, Emmanuel Suárez Acevedo, Shubh Agrawal, Joey Velez-Ginorio, and Stephanie Weirich

Awards

Graduate Research Fellowship, National Science Foundation (2024)

Outstanding Research Award, University of Pennsylvania SEAS (2024)

Dean's Excellence/Hopper-Dean Fellowship, Cornell Bowers CIS (2024)
